

CHINS 108

INFORMATION MANUAL

WARNING

DO NOT CLEAN, ADJUST OR REPAIR ANY MACHINERY WHILE PLUGGED INTO ELECTRICAL OUTLETS, KEEP HANDS AND ALL FOREIGN OBJECTS CLEAR OF MACHINERY WHILE IT IS IN OPERATION.

CUIDADO

NO LIMPIE, AJUSTE O REPARE CUALQUIER MAQUINARIA QUE ESTE ENCHUFADA. NO INTRODUCIR LAS MANOS O CUALQUIER OTRO OBJETO MIENTRAS LA MAQUINARIA ESTA EN ACCION.

ADJUSTMENTS TO COMBUSTION SYSTEMS

GENERAL INFORMATION

NOTE: AS YOU READ THE INFORMATION BELOW, REFER TO THE SYSTEM DIAGRAM (FIG. 1)

A. Casa Herrera uses a 'Proportional Mix' combustion system consisting of six (6) basic elements:

1. Combustion Air Blower
2. Actuator-operated Airflow Valve
3. Manually-adjustable Airflow Valve
4. Venturi-type Gas/Air Mixer/s
5. Fuel Gas Zero-Governor (pressure regulator valve)
6. Gas Burner(s)

OPERATION:

The **Combustion Air Blower (1)** generates pressurized air. The **Airflow Valve (2)**, operated by an actuator connected to a temperature control, meters out the amount of air needed for varying flame sizes. A **Manual Air Valve (3)** is factory-adjusted to limit the maximum possible flame size for each level. Air passing through the **Gas-Air Mixer (4)** causes a suction that draws Gas into the air-stream. The **Gas Zero Governor (5)** regulates gas pressure so that the resulting mixture is correct from High to Low flame.

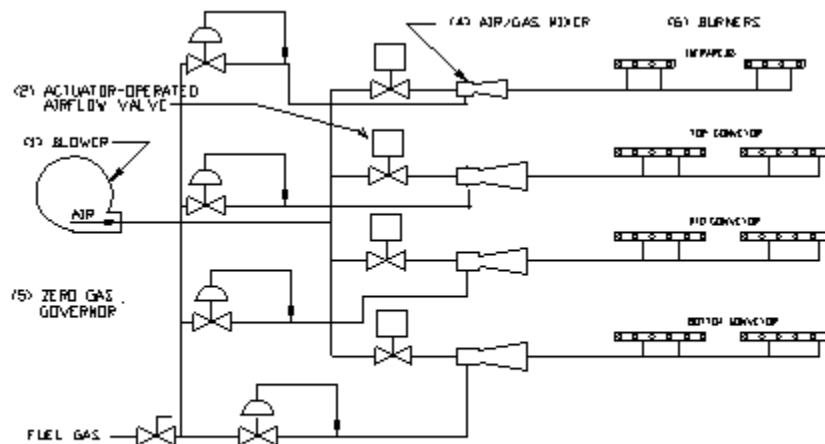


FIG. 1 SCHEMATIC OF PROPORTIONAL MIX COMBUSTION SYSTEM

ADJUSTMENTS TO COMBUSTION SYSTEMS

GENERAL INFORMATION (cont.)

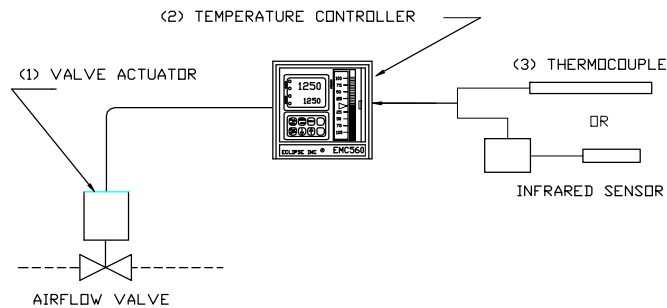


FIG. 2 SCHEMATIC OF THE TEMPERATURE CONTROL SYSTEM

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- B. A temperature Control System is also provided. It opens and closes the Airflow Valve to maintain a preset oven temperature. There are three basic parts to this system:

1. Valve Actuator
2. Temperature Controller
3. Thermocouple or other temp. sensor

When heat is applied to the Thermocouple (Item 3), a voltage is generated. The voltage level depends on how much heat. This voltage is measured by the controller (Item 2) and compared with the pre-set "setpoint" value.

A difference in voltage causes the controller to direct the Valve Actuator (Item 1) to open the Airflow Valve if the temperature is below setpoint, or close the valve if the temperature is above.

The valve actuator contains a position sensing and transmitting device. This tells the Controller what is happening at the valve and allows the controller to 'anticipate' changes and avoid large temperature oscillations.

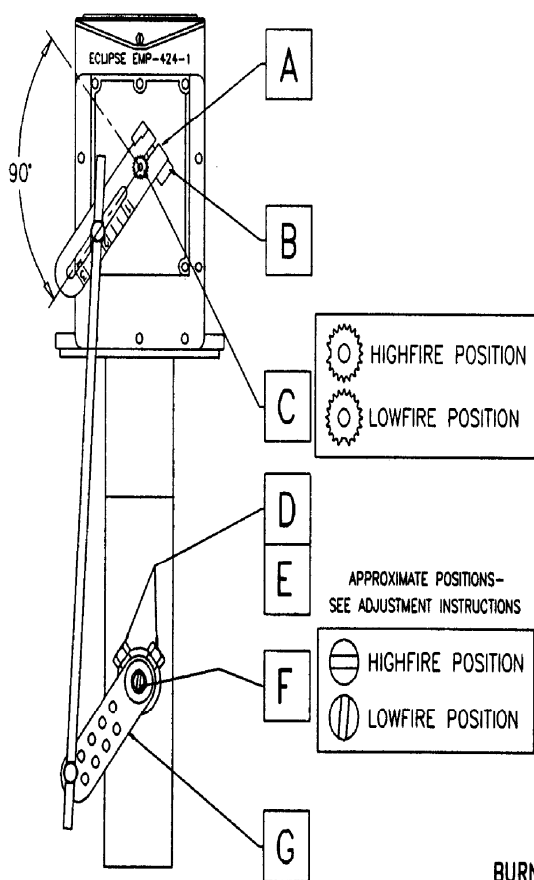
- C. SYSTEM ADJUSTMENTS:

See Part 2 for Main Air Valve
Part 3 for Mixers
Part 4 for Temperature Controllers

ADJUSTMENTS TO COMBUSTION SYSTEMS

Part 2. MAIN AIR VALVE AND ACTUATOR

TO RAISE TOTAL HEAT AVAILABLE (INCREASE FLAME HEIGHT) AT LOW FIRE SETTING.



1. a. Loosen CLAMP BOLT [B]
- b. Loosen LOCK SCREW [D]
- c. Loosen LOCK SCREW [E]
2. a. Adjust SET POINT of temperature controller to 0°.
- b. Check that shaft [C] is turning COUNTER-CLOCKWISE TO LOWFIRE.
3. a. Rotate ARM [A] to a 90° angle.
- b. Tighten CLAMP BOLT [B]
4. Close MAIN GAS COCK.
5. Start BLOWER.
6. a. Observe SHAFT [F] and note the LINE scribed across the end of the shaft. This line represents the position of the Butterfly Valve plate inside.
- b. Now turn SHAFT [F] clockwise until you get the LOWEST FLAME POSSIBLE.
- c. Tighten LOCKSCREWS [D] and [E]
7. Adjust SETPOINT of temperature controller to 500°.
8. a. Adjust SETPOINT of temperature controller to 0°.
- b. Check valve position.

BURNERS WILL GO OUT IF VALVE CLOSES COMPLETELY AT LOW-FIRE POSITION!

9. REPEAT INSTRUCTIONS 6, 7 AND 8 IF NECESSARY FOR PROPER VALVE ADJUSTMENT.

ADJUSTMENTS TO COMBUSTION SYSTEMS (CONT.)

Part 2. MAIN AIR VALVE AND ACTUATOR

- A.
 - 1. Mark position of TOP SWIVEL on valve ACTUATOR lever arm
 - 2. Mark position of BOTTOM SWIVEL on VALVE lever arm.
 - 3. Mark position of LINKAGE ROD at TOP and BOTTOM SWIVELS.

- B.
 - 1. Cycle oven to HIGH FIRE setting by turning temperature control knob CLOCKWISE.
 - 2. Check and RECORD pressure reading on large diameter VERTICAL AIR MANIFOLD.
 - 3. Check FLAME HEIGHT at all burners.

- C.
 - 1. Cycle oven to LOW FIRE setting by turning temperature control knob COUNTER CLOCKWISE.
 - 2. Check AND RECORD pressure reading at large diameter VERTICAL AIR MANIFOLD.
 - 3. Check FLAME HEIGHT at all burners.

- D.
 - 1. Check that oven is at LOW FIRE setting.
 - 2. Hold TOP SWIVEL and LINKAGE ROD.
 - 3. Carefully loosen BOLT securing LINKAGE ROD in SWIVEL.
 - 4. Slide LINKAGE ROD slowly upwards thru SWIVEL until pressure reading at VERTICAL AIR MANIFOLD increases by 1/4" w.c.
 - 5. Tighten bolt securing LINKAGE ROD in SWIVEL

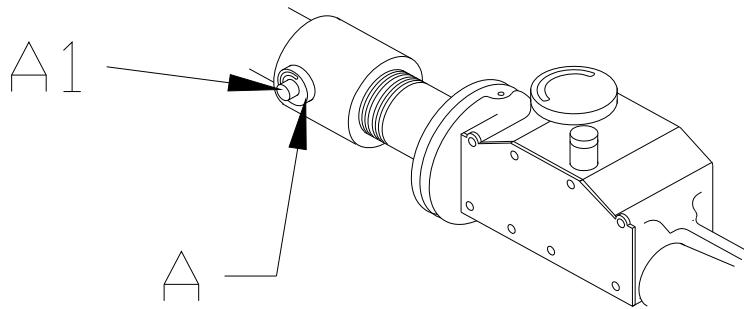
Part 2. MAIN AIR VALVE AND ACTUATOR

- E.
 - 1. Cycle oven to HIGH FIRE by turning temperature control knob CLOCKWISE.
 - 2. Hold TOP SWIVEL and valve ACTUATOR lever arm.
 - 3. Carefully loosen NUT securing SWIVEL to ACTUATOR lever arm
 - 4. Move SWIVEL along ACTUATOR lever arm TOWARDS ACTUATOR shaft until pressure reading at VERTICAL AIR MANIFOLD matches the reading recorded in STEP B.2.
 - 5. Tighten NUT securing SWIVEL to ACTUATOR- lever arm.

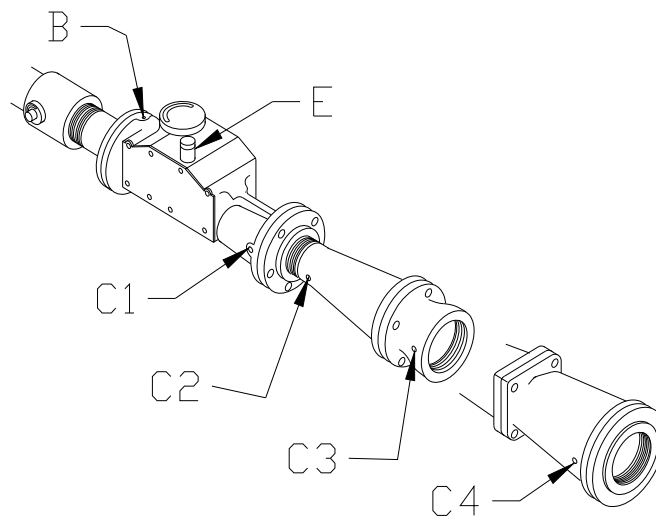
- F.
 - 1. REPEAT STEPS D and E until flame height is satisfactory at LOW FIRE, WITH NO CHANGE in pressure at VERTICAL AIR MANIFOLD at HIGH FIRE setting.

ADJUSTMENTS TO COMBUSTION SYSTEM MIXERS, 3 TIER OVEN

1. Close MAIN GAS COCK
2. Open MAIN AIR VALVE to high fire position by turning knob on temperature control to 750°.

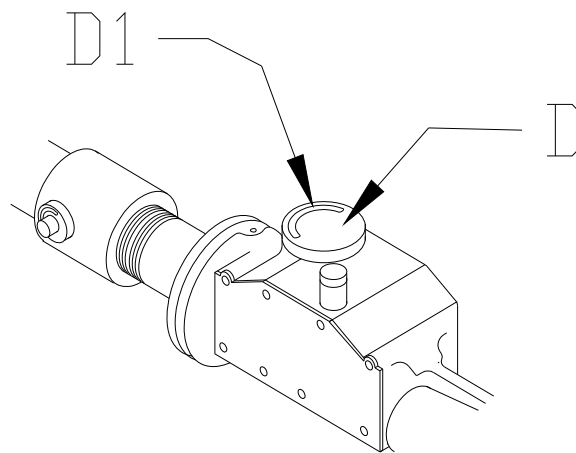


3.
 - a. Loosen thumbscrew A1
 - b. Open MIXER BUTTERFLY VALVE a full open.
 - c. Tighten thumbscrew A1 finger tight

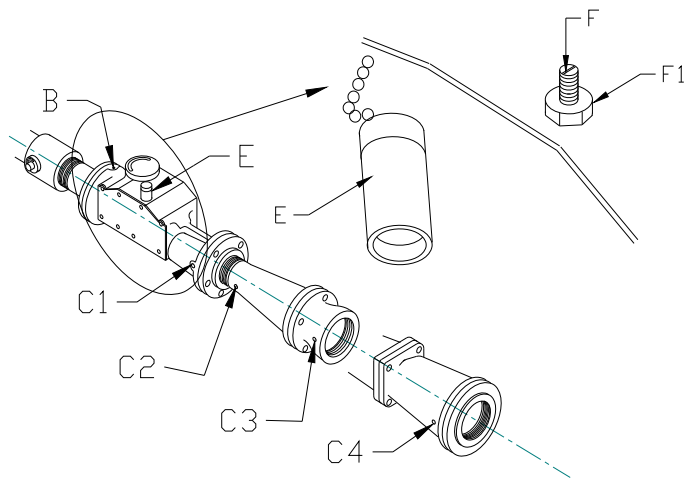


ADJUSTMENTS TO COMBUSTION SYSTEM MIXERS, 3 TIER OVEN

4. Install a 0-15" w c. gauge or manometer at B
5. Depending on the type of mixer used, install another gauge or manometer at C1, C2, C3, OR C4
6.
 - a. Loosen lock screw D1
 - b. Close FUEL SELECTOR D fully clockwise
 - c. Tighten lock screw D1



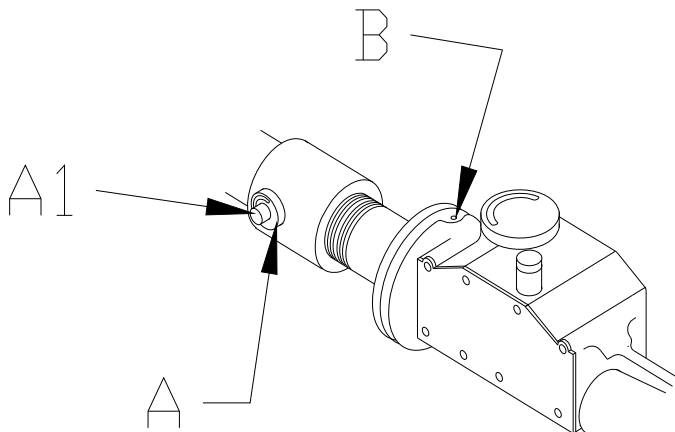
7. START BLOWER
8. Remove RATIO ADJUST COVER "E" by turning counter-clockwise and let hang by the security chain
9.
 - a. Loosen brass locknuts F1
 - b. Adjust RATIO SCREW in or out until pressure reading at C1, C2, C3, or C4 = $\frac{1}{4}$ of reading at B
 - c. Tighten locknut F1 finger tight



10. a. Loosen thumbscrews A1
- b. set MIXER BUTTERFLY VALVE A to give approximately 2-1/2 TO 4" W.C. press at B
- c. Tighten thumbscrew A1

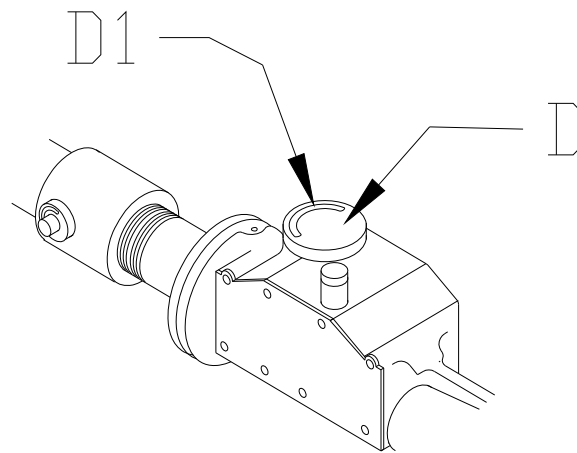
Supplementary Instructions

ADJUSTMENTS TO COMBUSTION SYSTEM MIXERS, 3 TIER OVEN



REPEAT INSTRUCTIONS 3 THRU 10 FOR EACH MIXER

11. Close MAIN AIR VALVE to low fire position by setting temperature control to 100°
12. CHECK AIR PRESSURE AT EACH MIXER AT TAP “B”. If pressure is below 1” w.c. refer to section 2, Adjustments to Main Air Valve
13. Open MAIN GAS COCK
14. Press reset button on Flame Control Relay Burner runner will ignite and Gas Valve open.

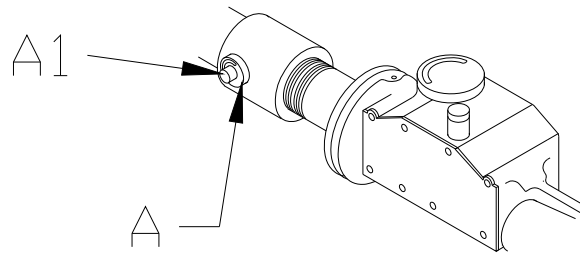


15.
 - a. Loosen lockscrew D1
 - b. Slowly Open FUEL SELECTOR D until burners ignite.
 - c. Adjust FUEL SELECTOR D to give a stable flame of proper color
 - d. Tighten lockscrew D1
16. Open AIR VALVE by setting temperature control to 750°

Supplementary Instructions

ADJUSTMENTS TO COMBUSTION SYSTEM MIXERS, 3 TIER OVEN

17. a. Loosen thumbscrews A1
- b. Adjust MIXER BUTTERFLY VALVE A for desired heat
- c. Tighten thumbscrew A1



REPEAT INSTRUCTIONS 15 THRU 17 FOR MIDDLE BURNER TIER & THEN TOP BURNER TIER

18. a. Cycle system Completely from high fire to low fire position several times by changing the set point on temperature control.

ALLOW SUFFICIENT TIME AT EACH SETTING FOR OVEN TO SETTLE DOWN.

- b. Check to see that all burners stay lit when system is at low fire.

MIXER RE-ADJUSTMENT FOR OTHER FUEL GASES

Before ovens are shipped from our manufacturing facility, the gas/air mixers are adjusted and calibrated to operate on 1000 BTU Natural Gas.

For operation on gases of different BTU content such as Butane or Propane, re-adjust the gas/air mixer Fuel Selector as detailed below.

19. a. Loosen lock screw D1
- b. Slowly Open FUEL SELECTOR D
- c. Adjust FUEL SELECTOR D in direction of Arrow to give a stable flame of proper color.
- d. Tighten lock screws E

